

Intel's Sandra Rivera on future of AI, data centers and India's tech moment

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“Anything anyone talks about right now is Generative AI and Large Language Models (LLMs), but AI is so much more than that,” emphasised Sandra Rivera of Intel. It's just the tip of the iceberg, the current cool kid on the technology block, the 'it' thing. In many ways it's the beginning of AI for a lot of people, but in no way the end of the subject.

Everything from data preparation that happens before training a machine learning model, its related data management, data filtering and cleaning stages, all of that also encapsulates the overall AI workflow. All of which is done almost exclusively upfront with a CPU, predominantly Intel's Xeon chips, according to Rivera. “Then you get to the model training phase, where you have small to medium sized models, we can address them very effectively with the CPU,” she said, explaining how any model that's 10 billion parameters or lower in size is very well suited for CPU workloads.

It's only when you talk about the largest models, with over a 100 billion parameters, where you need a parallel architecture, which is what a GPU provides with dedicated deep learning acceleration. “That's where we play with Intel's Habana Gaudi AI processors, where NVIDIA plays as well,” said Rivera. This is where the training and inference stage of an AI model workflow is built and validated.

“Bottomline to remember, AI is complex and multifaceted. Right now there's



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a lot of interest around these LLMs and Generative AI deployments in the cloud, we are seeing customers who want to have at least some GPU capability at a dynamic level. But in the long term, you know, the data center market is complex. I know people think it's all finished, but it's very early days of AI, we're just starting, and as Intel we think we have a very good opportunity to play with the breadth of capabilities we have across our portfolio,” summarised Rivera.

I met Sandra Rivera, Executive Vice President & General Manager, Data Center and AI Group, Intel Corporation, at a press briefing organised at Intel's India Development Center in Bengaluru, Karnataka. In a post-ChatGPT world,

where interest around Generative AI tools and applications is so high, Rivera explained how Intel's not only powering AI applications in the data center and cloud, but also preparing to challenge NVIDIA's supremacy in the market with respect to GPU-accelerated workloads.

“Everyone knows NVIDIA is doing a great job of delivering GPUs to the market and they're of course a giant player in the AI space, but we see that for customers CPU is still very much a big part of the equation,” because in many cases they already have an Intel Xeon CPU-powered business model, acknowledged Rivera. And the fact that they can do AI and machine learning workloads with the CPU also has opened a market opportunity and tailwind for Intel's data center business, according to her.

Intel, in case you don't know, is a powerhouse of the semiconductor industry, especially with CPU chips for both consumer and enterprise work. According to Counterpoint Research, Intel remained the leader in the data center CPU market with a 71% share in 2022. Whereas Nvidia enjoys more than a 90% share of the data center GPU market, according to IDC. While data centers are still largely dominated by CPUs compared to GPUs, the accelerated workloads share of the overall data center pie has grown faster than any other segment globally – all thanks to the rise of AI applications and workflows – which has led to NVIDIA's growth. As NVIDIA tries to diversify and attack Intel's overall data center supremacy, by trying to dip into the server CPU market, Intel is pulling up its socks and accelerating to not only protect their market share but also make inroads into NVIDIA's dominance into the GPU market. But where NVIDIA's H100 GPUs are very expensive, Intel's banking on providing a better overall value proposition (in terms of total cost of ownership) to customers looking for a strong alternative.

Intel's Habana Gaudi chips are powerful AI training processors, competing head on with NVIDIA's Tesla accelerators. Released in 2022, Intel's Gaudi2 chip is the only alternative to NVIDIA GPUs for